

Solver-Integrated Lossy and Lossless Compression for Scalable Flow Simulations

Viral Sudip Shah
University of Illinois Urbana-Champaign



THE STORAGE AND I/O WALL IN EXASCALE COMPUTING

Modern GPU-accelerated HPC simulations routinely generate datasets ranging from hundreds of gigabytes to **multiple terabytes** per snapshot, making storage, I/O, and data movement major bottlenecks in exascale scientific workflows. As compute capability continues to scale, filesystem bandwidth and storage infrastructure increasingly limit output cadence, visualization, and downstream analysis.

BOTTLENECKS

- Multi-terabyte simulation snapshots
- Filesystem and MPI-I/O bottlenecks
- High storage and data transfer costs
- Limited output cadence at extreme scale
- Expensive post-hoc analysis workflows

Result: I/O and storage wall limits science and productivity

OUR SOLUTION: SOLVER-INTEGRATED COMPRESSION

We integrate both **lossless** compression for restart/checkpoint reliability and **error-bounded lossy compression** for analysis-oriented outputs directly into the GPU-enabled spectral-element solver nekRS. By compressing solution fields in situ before MPI-I/O, we substantially reduce I/O volume and storage cost while preserving scientific value.

BENEFITS

- Reduces storage footprint and I/O volume
- Minimizes data movement overhead
- Preserves QoIs within controlled tolerances
- Enables higher output cadence at extreme scale
- Maintains scientific usability of reconstructed fields

Result: Lower I/O cost, higher output cadence, and scalable science

METHODOLOGY: COMPRESSION-AWARE I/O FRAMEWORK

A portable, compression-aware I/O framework integrated in CFD Solver



HPC Solver (nekRS)

- Spectral-element discretization
- High-order accuracy
- GPU-accelerated



Rank-local Fields (Element-based)

- Each rank holds a subset of elements
- No communication for compression



Compression (in situ)

- Error-bounded SZ3 for analysis
- Lossless Blosc2 for checkpoints



Portable Compressed Format (Self-describing)

- Rank-decomposable metadata
- Decouples storage from MPI layout
- Supports arbitrary read-time parallelism



Readers & Workflows (Coupled or Decoupled)

- Visualization · Post-hoc analysis
- ROM · ML / GNN pipelines
- In situ or ex situ access

QoI PRESERVATION

BOUNDS FROM POINTWISE TOLERANCE

Guaranteed pointwise error bound $|u_i - \tilde{u}_i| \leq \epsilon \cdot \Delta u_{range}$

Closed form a priori bounds on physical QoI

Pick ϵ from target accuracy

GENERALIZATION: A PRIORI BOUNDS FOR ANY QoI

Quantity of Interest (QoI)	Error-bound scaling
Velocity magnitude $ u $	$\leq \sqrt{3} \cdot \epsilon \cdot \Delta u_{range}$
Kinetic energy $E_k = \frac{1}{2} u ^2$	$O(\epsilon \cdot u_{max}^2)$
Turbulence kinetic energy k	$O(\epsilon \cdot u_{max}^2)$
Integral statistics $\langle \phi \rangle$	$O(\epsilon \cdot \phi)$
Gradients ∇u , vorticity ω	$O(\epsilon/h) \rightarrow \text{grid-dependent}$
Stress τ , Nusselt Nu	$O(\epsilon/h)$

WORKED EXAMPLE: FUSION BLANKET VELOCITY

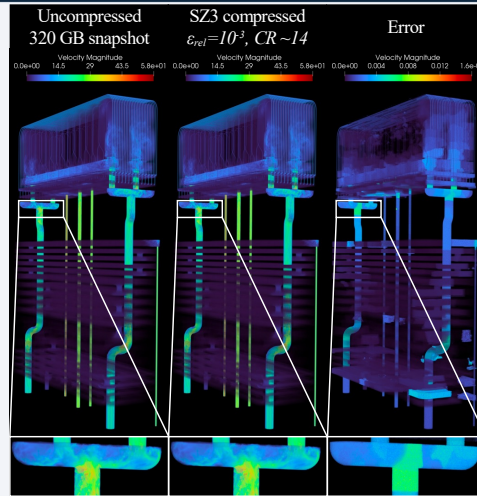
Given: $u_x, u_y, u_z \in [-58, 58]$, $\epsilon = 10^{-3}$, $\Delta u_{range} = 116$

Per-component bound: $|u_i - \tilde{u}_i| \leq \sqrt{3} \cdot \epsilon \cdot 0.116$

Magnitude bound: $||u| - |\tilde{u}|| \leq \sqrt{3} \cdot \epsilon \cdot \Delta u_{range} \approx 0.20$ (worst case)

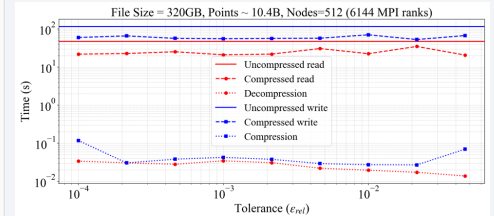
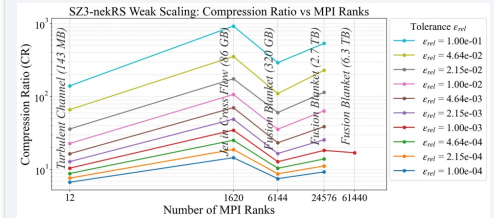
Observed: $||u| - |\tilde{u}|| \leq 1.6 \times 10^{-2} \rightarrow \sim 10\times$ tighter than worst-case

FUSION BLANKET: VELOCITY MAGNITUDE



Flow structures and features are preserved after compression.

PERFORMANCE: SCALING & I/O TIME



10-100x compression ratio and up to 8x I/O time reduction.

FUTURE WORK

- GPU-resident compression and asynchronous I/O
- SEM-aware transform-based compression strategies
- Integration with visualization tools (ParaView / VisIt)
- Evaluation of additional lossless and lossy compressors
- Additional directions:
- Reactive and Multiphysics flows
- Adaptive tolerance selection
- Compressed ML-ready datasets for ROM and AI workflows

GENERALIZATION

- Applicable to a wide range of distributed scientific workflows with locally structured data layouts.
- High-order FEM / DG
- Structured-grid CFD
- Multiphysics workflows
- Key ideas:
- MPI-rank-local compression
- In situ data reduction
- QoI-aware error-bounded compression
- Compressed parallel MPI-I/O

A general strategy for scalable, data-centric HPC workflows.

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Generative AI tools assisted with language editing and code/debugging tasks.

Data reduction is as critical as numerical computation in the exascale era.

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